```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns


%matplotlib inline

df = pd.read_csv("df_final_features.csv"
)

print("Dataset Shape:", df.shape)
display(df.head())
print(df.info())

Dataset Shape: (202616, 22)
```

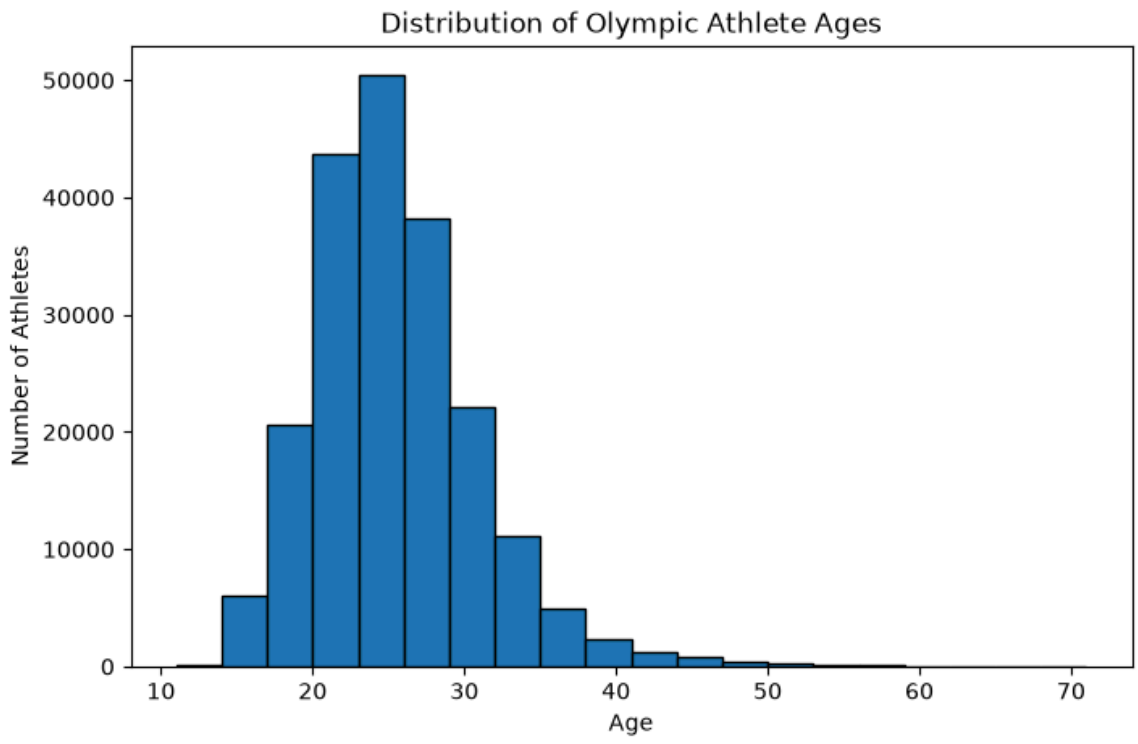
	Name	Sex	Age	Height	Weight	Team	Year	Season	Host_
0	A Dijiang	M	24.0	180.0	80.0	China	1992	Summer	Barce
1	A Lamusi	M	23.0	170.0	60.0	China	2012	Summer	Lond
2	Christine Jacoba Aaftink	F	21.0	185.0	82.0	Netherlands	1988	Winter	Calga
3	Christine Jacoba Aaftink	F	21.0	185.0	82.0	Netherlands	1988	Winter	Calga
4	Christine Jacoba Aaftink	F	25.0	185.0	82.0	Netherlands	1992	Winter	Alber

5 rows × 22 columns

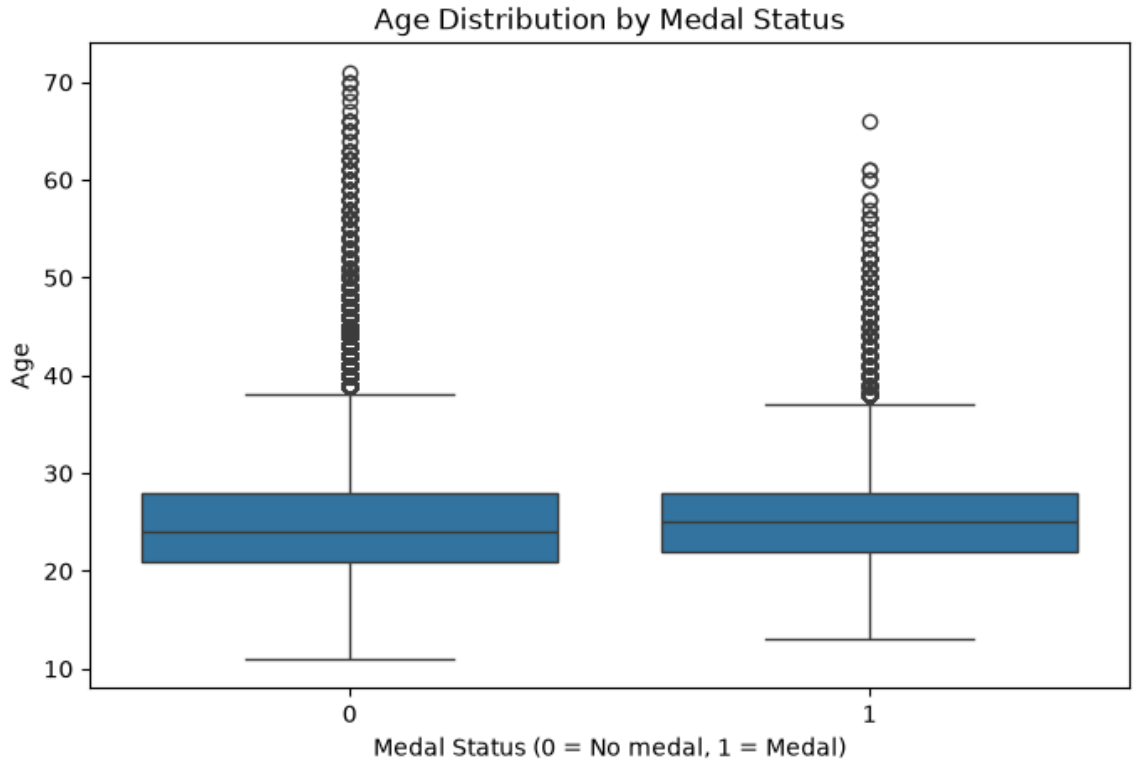
```
<class 'pandas.DataFrame'>
RangeIndex: 202616 entries, 0 to 202615
Data columns (total 22 columns):
 #   Column                                                  Non-Null Count   Dtype
---  ------                                                  --------------   -----
 0   Name                                                    202616 non-null  str
 1   Sex                                                     202616 non-null  str
 2   Age                                                     202616 non-null
float64
 3   Height                                                  202615 non-null
float64
 4   Weight                                                  202615 non-null
float64
 5   Team                                                    202616 non-null  str
 6   Year                                                    202616 non-null  int64
 7   Season                                                  202616 non-null  str
 8   Host_City                                               202616 non-null  str
 9   Host_Country                                            202616 non-null  str
 10  Sport                                                   202616 non-null  str
```

```
 11  Event                                          202616 non-null  str
 12  GDP_Per_Capita_Constant_LCU_Value              202616 non-null
float64
 13  Cereal_yield_kg_per_hectare_Value              202616 non-null
float64
 14  Military_expenditure_current_LCU_Value         202616 non-null
float64
 15  Tax_revenue_current_LCU_Value                  202616 non-null
float64
 16  Expense_current_LCU_Value                      202616 non-null
float64
 17  Central_government_debt_total_current_LCU_Value  202616 non-null
float64
 18  Representing_Host                              202616 non-null  int64
 19  Avg_Temp                                       202616 non-null  str
 20  Medal                                          202616 non-null  int64
 21  Medal_Binary                                   202616 non-null  int64
dtypes: float64(9), int64(4), str(9)
memory usage: 34.0 MB
None
```

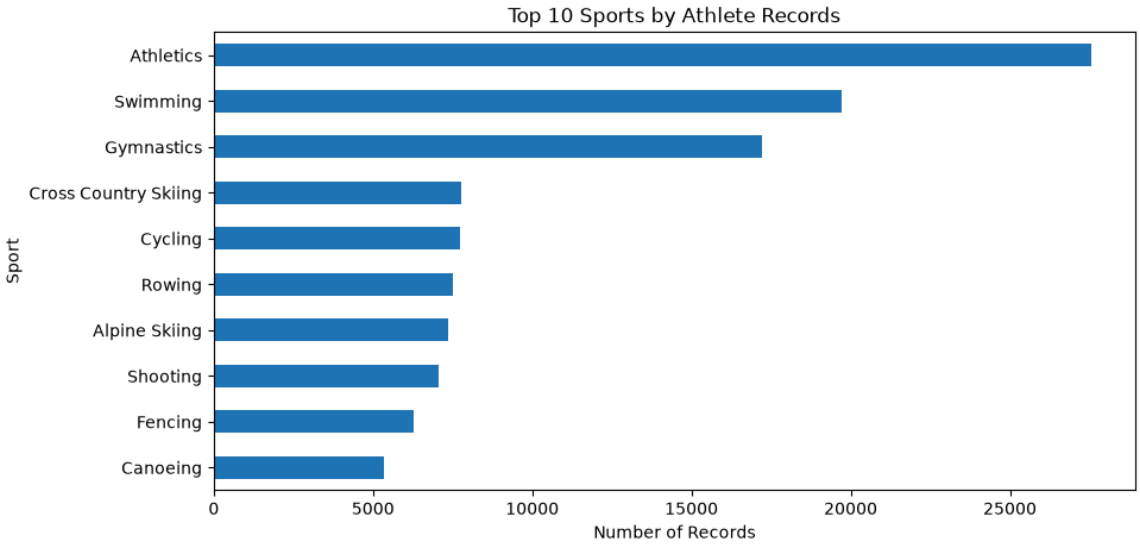
```
plt.figure(figsize=(8, 5))
plt.hist(df['Age'].dropna(), bins=20, edgecolor='black')
plt.title('Distribution of Olympic Athlete Ages')
plt.xlabel('Age')
plt.ylabel('Number of Athletes')
plt.show()
```



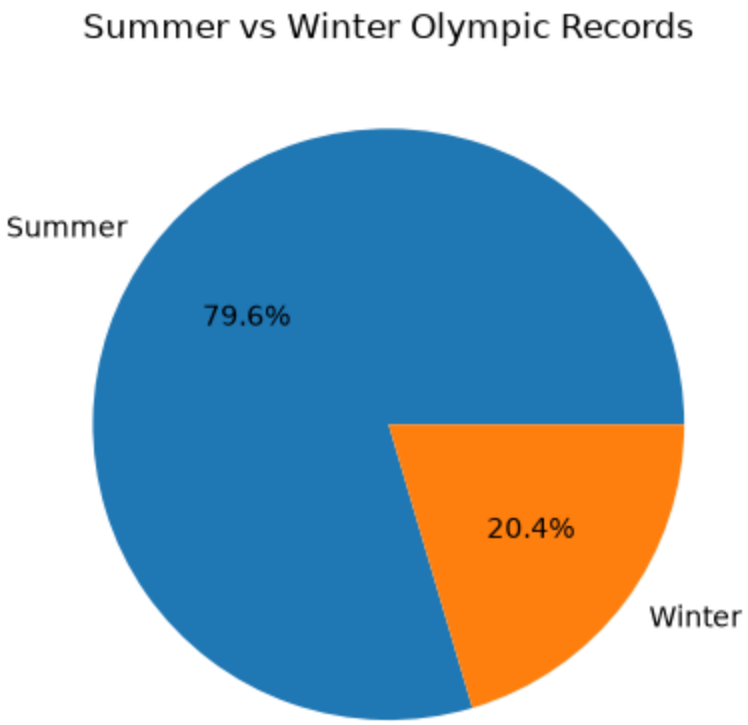

```
plt.figure(figsize=(8, 5))
sns.boxplot(data=df, x='Medal_Binary', y='Age')
plt.title('Age Distribution by Medal Status')
plt.xlabel('Medal Status (0 = No medal, 1 = Medal)')
plt.ylabel('Age')
plt.show()
```



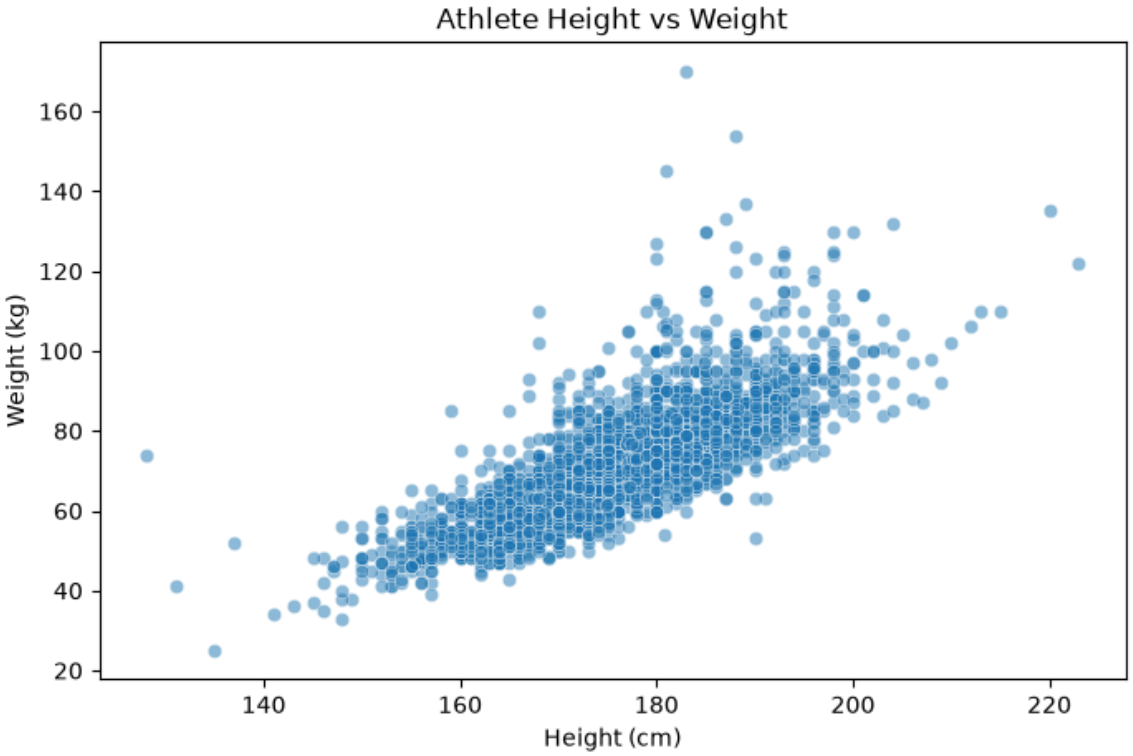

```
plt.figure(figsize=(10, 5))
df['Sport'].value_counts().head(10).sort_values().plot(
    kind='barh'
)
plt.title('Top 10 Sports by Athlete Records')
plt.xlabel('Number of Records')
plt.ylabel('Sport')
plt.show()
```




```
df['Season'].value_counts().plot(
    kind='pie', autopct='%1.1f%%'
)
plt.title('Summer vs Winter Olympic Records')
plt.ylabel('')
plt.show()
```




```
plt.figure(figsize=(8, 5))
sns.scatterplot(
    data=df.sample(n=min(3000, len(df)), random_state=42),
    x='Height',
    y='Weight',
    alpha=0.5
)
plt.title('Athlete Height vs Weight')
plt.xlabel('Height (cm)')
plt.ylabel('Weight (kg)')
plt.show()
```




```
medals = (
    df[df['Medal_Binary'] == 1]
    .groupby('Year')
    .size()
    .sort_index()
)

plt.figure(figsize=(10, 5))
plt.plot(medals.index, medals.values, marker='o')
plt.title('Medal-Winning Records by Year')
plt.xlabel('Olympic Year')
plt.ylabel('Number of Records')
plt.grid(True)
plt.show()
```

